

NCSA/PSA Open Hackathon

LBL ATLAS

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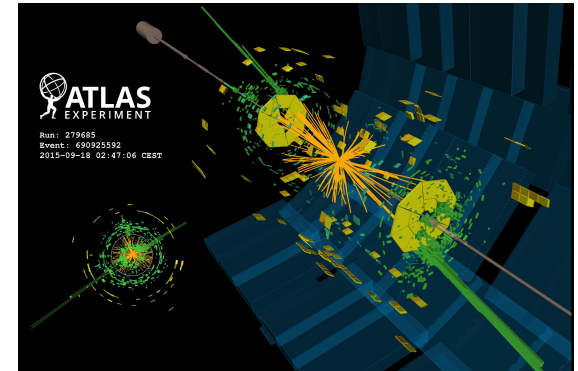
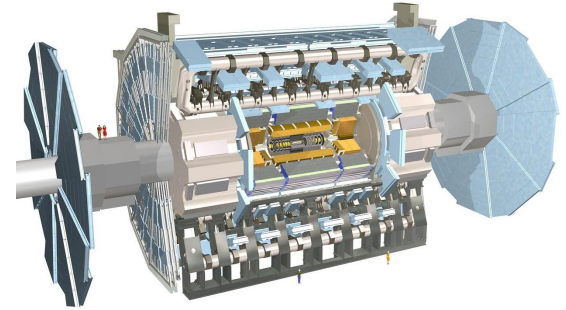


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The ATLAS Experiment

- ATLAS is a particle detector to capture data from proton-proton collisions at the Large Hadron Collider
- We use this data to study fundamental particles and forces, measure Standard Model processes, and search for new physics
- Turning collision data into a physics result requires complex analysis workflows: event selection, background modeling, statistical fitting, and uncertainty evaluation
- We aim to help small language models assist with these analysis workflows via post-training



Application Details

Base Model: Qwen2.5-Instruct, 1.5B and 7B parameters

Goal: Specialize compact models for HEP reasoning and analysis tool use without full model fine-tuning

Training Framework: [ByteDance VERL](#)

- LoRA: to preserve the model's original intelligence
- FSDP: train effectively on 4 GPUs (Nvidia A100s)

Dataset Details

Dataset 1: [HEP Signature Backgrounds](#)

Training Objective: HEP background composition knowledge

Input: Reconstructed object multiplicities: leptons, photons, jets, b-tags, MET

Task: identify the proper background process given a signal region definition

Scale: 8037/240/297 Train/Val/Test

Dataset 2: [HEP Config SFT](#)

Training Objective: Tool-using analysis configuration competence

Input: Natural language task + starting JSON analysis config + tool schemas

Task: Use tool calls to modify the starting config

Scale: 2400/360/600 Train/Val/Test

Hackathon Goals

1. Create an executable TRExFitter training environment
 - There is this package called [TRExFitter](#), that reduces an end-to-end HEP analysis to a single [yaml config](#)
 - Our goal is to generate SFT/RL examples in which agents create, edit, validate, and run TRExFitter analysis configurations
 - Agents are scored for correctness and executability, then for physics performance
2. Establish an HEP agent post-training workflow
 - Current we are using [VERL](#) out of the box
 - Are there any other training packages we should look into for SFT and RL?
3. Convert Harbor evaluations into training data
 - We have been using Harbor to benchmark agent performance on several different physics tasks
 - Mine Harbor agent trajectories into SFT/RL training dataset